Raghav Gupta

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Machine Learning Scientist with 4+ years across deep learning research and production ML. Working on multimodal deep learning, vision-language-action models, VLM mid-training and fine-tuning, diffusion generation, and LLM agents.

Experience

Fast Retailing (UNIQLO) Tokyo, Japan
Machine Learning Scientist *Aug 2021 – Jun 2025*

- Rebuilt SKU-level demand forecasting as a production system, replacing planner heuristics with hierarchical and probabilistic models over lag, seasonality, and holiday signals. Cut weighted absolute percentage error (WAPE) on the existing catalog from **33%→30%**, saving an estimated **¥3B** through better inventory optimization.
- Extended forecasting to newly launched items, which have no sales history, by learning multimodal product embeddings that transfer demand from visually and textually similar products. Feeding automated SKU ordering, this contributed to a further **¥6B+ reduction in excess inventory**.
- Built LLM-driven analytics and text-to-SQL tooling so merchandising teams could query sales and inventory data directly, eliminating **100+ hours** of manual analysis.

Education

Carnegie Mellon University Pittsburgh, PA
M.S. in Machine Learning and Data Science, GPA: 4.0/4.0 *Aug 2025 – Dec 2026*
Coursework: Generative AI, Deep Learning Systems, AI Agents, Multimodal Machine Learning

Indraprastha Institute of Information Technology Delhi Delhi, India
B.Tech. in Computer Science, GPA: 9.21/10.00 *Aug 2017 – Jun 2021*

Publications

VLAFactory: Data-Centric Pipeline for VLA Model Training [code] *IROS 2026 ScaleInfra (accepted)*

- Owned the **mid-training** stage of an open-source **vision-language-action** pipeline, comparing six data recipes and three adaptation strategies (**full fine-tuning, LoRA, QLoRA**) to test how training before control shapes a driving model.
- Carried every checkpoint through general, embodied, and closed-loop driving evaluation. Mid-training lifted **VLADBench 16.9→40.9** and **Bench2Drive 83.3→86.6** (success **60.9%→68.1%**), but intermediate VLM gains did not reliably predict downstream driving performance.

Selected Projects

- **Asynchronous Multi-Turn RL for LLM Agents**: Built a reinforcement learning framework that trains LLM agents to use tools across multi-turn tasks, with REINFORCE, GRPO, and PPO-style updates. Its asynchronous GRPO trainer overlaps rollout generation with learning across 256 concurrent episodes, dropping rollouts from outdated policies and correcting the rest with importance weighting to keep training stable. Trained Qwen3.5-4B with LoRA on self-hosted H100s, using Megatron-LM for training and vLLM for sampling.
- **Twitter Analytics Microservice on AWS**: Built a user-recommendation web service over 1 TB of raw tweets: a PySpark ETL pipeline on GCP Dataproc loads 40.8M rows into MySQL, served by a Go and gRPC microservice on Kubernetes, deployed with Terraform, Helm, and GitHub Actions CI/CD. Redesigning the schema to avoid joins raised throughput **6.8**×. Moved it to AWS managed services, choosing EKS with RDS MySQL over Fargate and DynamoDB after benchmarking all three; it sustained **7,411 RPS** at $0.45/hour.
- **Agent Harness for Coding and Game-Playing LLM Agents**: Built a ReAct agent harness from scratch with tool calling, step budgets, and sandboxed code execution, driving a repository bug-fixing agent and a game-playing agent. Context compaction halved the peak prompt but netted only 8% savings after counting summarizer cost and redone work.
- **GoT-Diffuse: Compositional Text-to-Image Generation** [report]: Asked why reasoning-based text-to-image models misplace objects even when their layout plans are correct: on T2I-CompBench++ plans are spatially valid 83.1% of the time but only 64.7% of images satisfy them. Layout-guided conditioning lifted spatial relations 19% and color binding 15%.
- **A PyTorch-like Deep Learning Framework**: Built reverse-mode autodiff, neural modules, and optimizers from scratch, plus tiled GPU matmul kernels and a FlashAttention-style optimization cutting attention memory 20%+.

Technical Skills

Languages: Python, Go, SQL, C++, CUDA
Frameworks/Tools: PyTorch, Scikit-learn, LightGBM, LangChain/LangGraph, NumPy, Pandas, BigQuery, Vertex AI, GCP, Modal, Docker, Kubernetes, Helm, Terraform, FastAPI, MLflow, Git, Weights & Biases
Core Areas: Deep Learning, Generative AI, Multimodal Learning, Reinforcement Learning, LLM Agents, RAG, Time-Series Forecasting, Parameter-Efficient Fine-Tuning